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2027 CYCLE

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CUET UG

Mathematics

# Vectors and Three-Dimensional Geometry

Products, Lines, Planes and Distances · Section B1 IV · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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## Syllabus Alignment

| Sub-topic | Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline) | Skills the questions test |
|---|---|---|
| **Vector algebra and types** | "Vectors and Scalars, magnitude and direction of a vector, Direction cosines and direction ratios of a vector. Types of vectors (equal, unit, zero, parallel and collinear vectors), position vector of a point, negative of a vector, components of a vector, addition of vectors, multiplication of a vector by a scalar, position vector of a point dividing a line segment in a given ratio." (Section B1 Mathematics Unit IV) | Component arithmetic at speed; section formula; l, m, n discipline. |
| **The three products** | "Definition, Geometrical Interpretation, properties and application of scalar (dot) product of vectors, vector (cross) product of vectors, scalar triple product of vectors." (Section B1 Mathematics Unit IV) | Angle, perpendicularity, area and volume computations with the right product. |
| **Lines in space** | "Cartesian equation and vector equation of a line, coplanar and skew lines, shortest distance between two lines. Direction cosines and direction ratios of a line joining two points." (Section B1 Mathematics Unit IV) | Writing line equations both ways; classifying skew/parallel/intersecting; SD. |
| **Planes and distances** | "Cartesian and vector equation of a plane. Angle between (i) two lines, (ii) two planes, (iii) a line and a plane. Distance of a point from a plane." (Section B1 Mathematics Unit IV) | Normal-driven plane writing; the three angle formulas; point-plane distance. |

**EXAM PATTERN**

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The 2026 syllabus houses this unit as **Section B1 Mathematics Unit IV** — the most formula-dense unit of the paper, which makes the "which product / which formula" reflexes on the following pages the real exam skill.

## Vector Basics — Magnitude, Components, Section Formula

A **vector** carries magnitude **and** direction; a **scalar** is a bare number. In components, $\vec{a} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$ with magnitude $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$. The type-names the syllabus lists are all quick checks: a **unit** vector has magnitude 1 (divide any non-zero vector by its length to make one), the **zero** vector has magnitude 0 and no defined direction, two vectors are **equal** when components match, and **parallel (collinear)** vectors are scalar multiples of each other. Addition runs component-wise, with the triangle law as its picture: walk $\vec{a}$ then $\vec{b}$, and the shortcut from start to finish is $\vec{a} + \vec{b}$.

The **section formula** locates a point dividing the join of $A$ (position vector $\vec{a}$) and $B$ (position vector $\vec{b}$) in ratio $m : n$:

$$\text{internal:} \quad \vec{r} = \frac{m\vec{b} + n\vec{a}}{m + n} \qquad \text{external:} \quad \vec{r} = \frac{m\vec{b} - n\vec{a}}{m - n}$$

with the midpoint as the $1 : 1$ special case, $\vec{r} = \frac{\vec{a}+\vec{b}}{2}$. Weight tilts toward the heavier letter — the point dividing in $2 : 1$ sits twice as close to $B$ as to $A$.

**WORKED EXAMPLE — RATIO SPLIT WITHOUT A SKETCH**

Find the position vector of the point dividing the join of $A\left(2\hat{\imath} + \hat{\jmath} - \hat{k}\right)$ and $B\left(4\hat{\imath} + 5\hat{\jmath} + 3\hat{k}\right)$ internally in the ratio $1 : 3$, and the midpoint of the same segment.

**Solution.** Internal, $m = 1$, $n = 3$:

$$\vec{r} = \frac{1\vec{b} + 3\vec{a}}{1 + 3} = \frac{(4 + 6)\hat{\imath} + (5 + 3)\hat{\jmath} + (3 - 3)\hat{k}}{4} = \frac{5}{2}\hat{\imath} + 2\hat{\jmath}.$$

Midpoint: $\frac{\vec{a}+\vec{b}}{2} = \frac{6\hat{\imath}+6\hat{\jmath}+2\hat{k}}{2} = 3\hat{\imath} + 3\hat{\jmath} + \hat{k}$. Notice the ratio point sits much closer to $A$ — the "heavier letter" check passes.

## Direction Cosines and Direction Ratios

A vector's direction is measured by the **direction cosines** $l, m, n$ — the cosines of the angles it makes with the $x$, $y$ and $z$ axes. For a vector with direction ratios $a, b, c$ (any proportional triple):

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

and the policing identity $l^2 + m^2 + n^2 = 1$ holds **always**. Direction **ratios** only need to be proportional — $(2, -4, 6)$ and $(1, -2, 3)$ describe the same direction — while direction **cosines** are the normalised version carrying the identity. For the line joining $P(x_1, y_1, z_1)$ to $Q(x_2, y_2, z_2)$, the direction ratios are simply the differences $x_2 - x_1$, $y_2 - y_1$, $z_2 - z_1$.

**WORKED EXAMPLE — RATIOS TO COSINES**

A line has direction ratios $(2, -1, -2)$. Find its direction cosines and verify the identity.

**Solution.** Magnitude: $\sqrt{4 + 1 + 4} = 3$. So $l = \frac{2}{3}$, $m = -\frac{1}{3}$, $n = -\frac{2}{3}$. Check: $l^2 + m^2 + n^2 = \frac{4}{9} + \frac{1}{9} + \frac{4}{9} = 1$ ✓. The signs carry the direction — the vector points along $+x$ and downward.

If an option's triple of "cosines" squares to something other than 1, it is a ratio triple wearing a cosine costume.

## The Dot Product — the Number That Measures Alignment

The scalar product fuses two vectors into one number with a geometrical meaning built in:

$$\vec{a} \cdot \vec{b} = |\vec{a}|\left|\vec{b}\right| \cos\theta = a_1 b_1 + a_2 b_2 + a_3 b_3$$

It answers "how much do these two point the same way?". Positive: acute angle. Zero: **perpendicular** — the single most used fact in the chapter, since every perpendicularity test in 3D geometry is a dot product. Negative: obtuse. The **projection** of $\vec{a}$ on $\vec{b}$ is $\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$ — the shadow $\vec{a}$ casts on $\vec{b}$'s direction.

**WORKED EXAMPLE — ANGLE FROM THE DOT**

Find the angle between $\vec{a} = \hat{\imath} + 2\hat{\jmath} + 2\hat{k}$ and $\vec{b} = 2\hat{\imath} + \hat{\jmath} - 2\hat{k}$.

**Solution.** Magnitudes: $|\vec{a}| = \sqrt{1+4+4} = 3$, $\left|\vec{b}\right| = \sqrt{4+1+4} = 3$. Dot: $1\cdot 2 + 2\cdot 1 + 2\cdot(-2) = 2 + 2 - 4 = 0$. Dot zero means $\theta = 90°$ — the vectors are perpendicular, and no cosine needs evaluating. Spotting the zero dot **is** the technique; computing further wastes the gift.

## The Cross Product — the Vector That Measures Turning

The vector product $\vec{a} \times \vec{b}$ produces a **vector** perpendicular to both inputs, with magnitude

$$\left|\vec{a} \times \vec{b}\right| = |\vec{a}|\left|\vec{b}\right| \sin\theta$$

and direction fixed by the right-hand rule. Two consequences do all the exam work: first, $\vec{a} \times \vec{b} = \vec{0}$ exactly when the vectors are **parallel** — the partner fact to "dot zero means perpendicular"; second, $\left|\vec{a} \times \vec{b}\right|$ **is** the area of the parallelogram with sides $\vec{a}, \vec{b}$, so a triangle built on them has area $\frac{1}{2}\left|\vec{a} \times \vec{b}\right|$. Computationally, expand the symbolic determinant with $\hat{\imath}, \hat{\jmath}, \hat{k}$ along the top row. The product **anti-commutes**: $\vec{a} \times \vec{b} = -\left(\vec{b} \times \vec{a}\right)$ — order flips sign, and the MCQ options know it.

**WORKED EXAMPLE — AREA OF A TRIANGLE FROM A CROSS**

Find the area of the parallelogram whose adjacent sides are $\vec{a} = \hat{\imath} - 2\hat{\jmath} + 3\hat{k}$ and $\vec{b} = 2\hat{\imath} + \hat{\jmath} - \hat{k}$.

**Solution.** Cross product by determinant expansion:

$$\vec{a} \times \vec{b} = \hat{\imath}((-2)(-1) - 3\cdot 1) - \hat{\jmath}(1\cdot(-1) - 3\cdot 2) + \hat{k}(1\cdot 1 - (-2)\cdot 2) = -\hat{\imath} + 7\hat{\jmath} + 5\hat{k}.$$

Magnitude: $\sqrt{1+49+25} = \sqrt{75} = 5\sqrt{3}$. Parallelogram area $= 5\sqrt{3}$ square units; a triangle on the same sides would halve it to $\frac{5\sqrt{3}}{2}$. One determinant, one square root, done.

## The Scalar Triple Product — Volume and Coplanarity

Stack the three products: $\left[\vec{a}\vec{b}\vec{c}\right] = \vec{a} \cdot \left(\vec{b} \times \vec{c}\right)$ — a **number** equal to the determinant of the $3 \times 3$ matrix of components. Its geometric gift: $\left|\left[\vec{a}\vec{b}\vec{c}\right]\right|$ is the **volume of the parallelepiped** with the three edges, and a **tetrahedron** built on them has volume $\frac{1}{6}$ of that. The algebraic gift: the triple product is **zero exactly when the three vectors are coplanar** — the flatness test that decides whether three points with a fourth form a proper solid or a squashed sheet.

**WORKED EXAMPLE — ARE THEY COPLANAR?**

Show that $\vec{a} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}$, $\vec{b} = 2\hat{\imath} + \hat{\jmath}$, $\vec{c} = \hat{\jmath} + \hat{k}$ lie in one plane.

**Solution.** Determinant rows $(1, 2, 3)$, $(2, 1, 0)$, $(0, 1, 1)$:

$$\left[\vec{a}\vec{b}\vec{c}\right] = 1(1 \cdot 1 - 0 \cdot 1) - 2(2 \cdot 1 - 0 \cdot 0) + 3(2 \cdot 1 - 1 \cdot 0) = 1 - 4 + 6 = 3 \neq 0.$$

The value is 3 — **not** zero, so these three are actually NOT coplanar; they span a parallelepiped of volume 3. Had the expansion hit zero, coplanarity would be declared. The lesson: the determinant decides, and a non-zero answer is a verdict too — read the question's demand before concluding.

## Lines in Space — Equations, Skew Lines, Shortest Distance

A line is pinned by **a point and a direction**. Vector form: $\vec{r} = \vec{a} + \lambda\vec{b}$ — start at $\vec{a}$, walk along $\vec{b}$. Cartesian form splits it per coordinate:

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

where $(l, m, n)$ are direction ratios. Two lines in space can **intersect** (a common point), be **parallel** (same direction, no meeting), or — the 3D novelty — be **skew**: not parallel yet never meeting, living on different altitudes. The **shortest distance** between skew lines with points $\vec{a}_1, \vec{a}_2$ and directions $\vec{b}_1, \vec{b}_2$ is

$$\text{SD} = \frac{\left|(\vec{a}_2 - \vec{a}_1) \cdot \left(\vec{b}_1 \times \vec{b}_2\right)\right|}{\left|\vec{b}_1 \times \vec{b}_2\right|}$$

— the common-perpendicular segment, measured by projecting the join onto $\vec{b}_1 \times \vec{b}_2$. For **parallel** lines the cross product vanishes, so the formula switches to $\frac{\left|(\vec{a}_2 - \vec{a}_1) \times \vec{b}\right|}{\left|\vec{b}\right|}$. Classify first (are the direction vectors proportional?), then pick the formula — the wrong-parallel-formula option is a standing trap.

**WORKED EXAMPLE — SHORTEST DISTANCE, END TO END**

Find the shortest distance between the lines $\vec{r} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k} + \lambda\left(\hat{\imath} - \hat{\jmath} + \hat{k}\right)$ and $\vec{r} = 2\hat{\imath} - \hat{\jmath} - \hat{k} + \mu\left(2\hat{\imath} + \hat{\jmath} - 2\hat{k}\right)$.

**Solution.** Directions $(1,-1,1)$ and $(2,1,-2)$ are not proportional — and the lines do not intersect, so they are **skew**. Cross: $\vec{b}_1 \times \vec{b}_2 = \hat{\imath}(1-1) - \hat{\jmath}(-2-2) + \hat{k}(1+2) = 4\hat{\jmath} + 3\hat{k}$, magnitude 5. Join: $\vec{a}_2 - \vec{a}_1 = \hat{\imath} - 3\hat{\jmath} - 4\hat{k}$. Dot: $0 - 12 - 12 = -24$.

$$\text{SD} = \frac{|-24|}{5} = \frac{24}{5} = 4.8 \text{ units.}$$

Classify, cross, project — three moves, one clean answer.

## Planes — Normals, Angles, and the Point-Plane Distance

A plane's identity card is its **normal** — a vector perpendicular to every line lying in it. Point-normal form: through $\vec{a}$ with normal $\vec{n}$, the plane is $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$, which in Cartesian clothes reads $ax + by + cz = d$ with $(a, b, c)$ the normal's direction ratios. Every angle question is a normal-or-direction dot product in disguise:

| Angle between | Formula | Note the trig |
|---|---|---|
| Two lines | $\cos\theta = \frac{\lvert\vec{b}_1 \cdot \vec{b}_2\rvert}{\lvert\vec{b}_1\rvert\lvert\vec{b}_2\rvert}$ | cosine of directions |
| Two planes | $\cos\theta = \frac{\lvert\vec{n}_1 \cdot \vec{n}_2\rvert}{\lvert\vec{n}_1\rvert\lvert\vec{n}_2\rvert}$ | cosine of normals |
| Line and plane | $\sin\theta = \frac{\lvert\vec{b} \cdot \vec{n}\rvert}{\lvert\vec{b}\rvert\lvert\vec{n}\rvert}$ | **sine** — the line tilts away from the normal |

The **distance of a point** $(x_1, y_1, z_1)$ from the plane $ax + by + cz = d$ is

$$\text{distance} = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

— substitute, subtract, divide by the normal's length. The absolute value makes sign irrelevant, which is why options differing only in sign are pure bait.

**WORKED EXAMPLE — DISTANCE AND A PLANE'S ANGLE**

Find the distance of the point $(1,-2,4)$ from the plane $2x + y - 2z = 5$, and the angle between this plane and the plane $x + 2y + 2z = 3$.

**Solution.** Distance: $\frac{|2\cdot1+(-2)-2\cdot4-5|}{\sqrt{4+1+4}} = \frac{|2-2-8-5|}{3} = \frac{13}{3}$ units. Angle: normals $(2,1,-2)$ and $(1,2,2)$; dot $= 2 + 2 - 4 = 0$ — the normals are perpendicular, so the planes meet at 90°. A zero dot between normals is the plane-perpendicularity signal; no cosine table required.

**EVERYDAY EXAMPLE — SUNLIGHT, SHADOWS AND TRIPODS**

The dot product is **sunlight geometry**: how directly the sun hits a solar panel is the panel's normal dotted with the sun's rays — full power when they align, zero at noon-equinox edges when they are perpendicular. The cross product is a **tripod's stability**: the further your feet spread (bigger $\sin\theta$), the larger the swept area that keeps you upright — a zero cross product means your feet are in a line and you topple. The triple product is a **moving box**: three edges

span a real box only if they point out of a common sheet; flatten them onto one table and the volume — the triple product — collapses to zero.

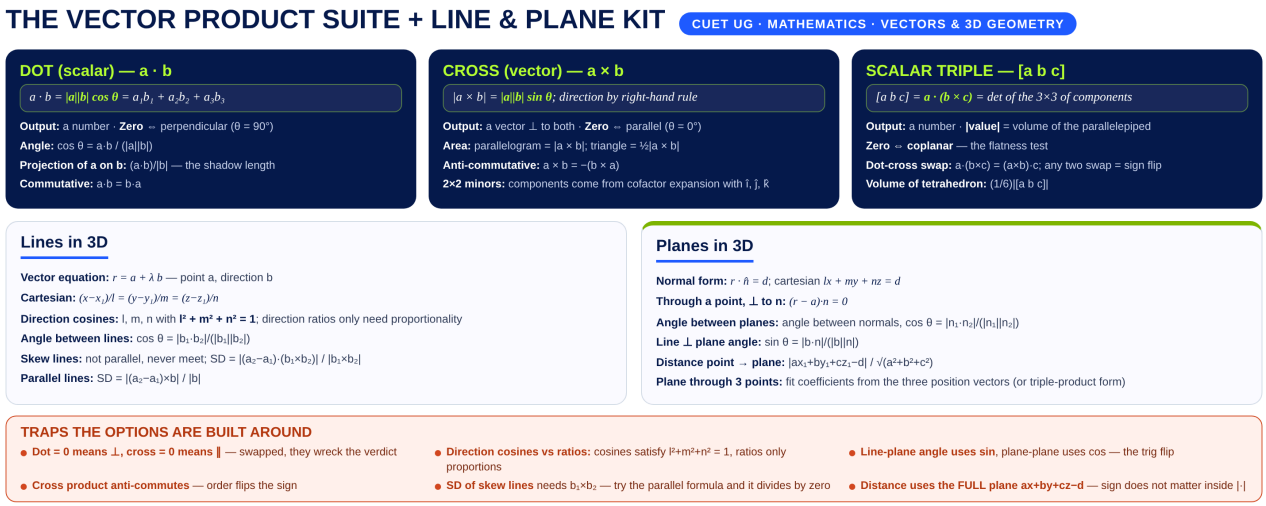



Figure 1: The product suite and the line-plane kit: three dark cards for the products, the line and plane formula shelves, and the traps under each.

## Formula Cheatsheet — the Whole Chapter on One Page

**FORMULA CHEATSHEET — VECTORS AND 3D GEOMETRY**

| Formula / rule | Letters mean | Fire it when |
|---|---|---|
| $\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ | Pythagoras in 3D | any magnitude |
| Unit vector: $\frac{\vec{a}}{\|\vec{a}\|}$ | length 1 | direction questions |
| Section: $\frac{m\vec{b}+n\vec{a}}{m+n}$ | internal division | ratio points |
| Midpoint: $\frac{\vec{a}+\vec{b}}{2}$ | the 1 : 1 case | quick checks |
| Direction cosines: $l^2 + m^2 + n^2 = 1$ | normalised ratios | cosine conversions |
| $\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\| \cos\theta$ | alignment meter | angles, projections |
| $\vec{a} \cdot \vec{b} = 0 \Leftrightarrow$ perpendicular | the perpendicular test | all 3D right angles |

| Formula / rule | Letters mean | Fire it when |
|---|---|---|
| Projection: $\frac{\vec{a}\cdot\vec{b}}{\lvert\vec{b}\rvert}$ | shadow length | component along a direction |
| $\lvert\vec{a}\times\vec{b}\rvert = \lvert\vec{a}\rvert\lvert\vec{b}\rvert\sin\theta$ | turning meter | parallel tests, areas |
| Parallelogram area $\lvert\vec{a}\times\vec{b}\rvert$; triangle $\frac{1}{2}$ of it | sweep area | area questions |
| $[\vec{a}\vec{b}\vec{c}] = \vec{a}\cdot(\vec{b}\times\vec{c})$ | 3 x 3 determinant | volumes, coplanarity |
| Volume: parallelepiped $\lvert\text{STP}\rvert$, tetrahedron $\frac{1}{6}\lvert\text{STP}\rvert$ | absolute value | solid questions |
| Line: $\vec{r} = \vec{a} + \lambda\vec{b}$; cartesian twin | point + direction | line equations |
| Skew SD: $\frac{\lvert(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)\rvert}{\lvert\vec{b}_1\times\vec{b}_2\rvert}$ | project on common perpendicular | skew pairs |
| Parallel SD: $\frac{\lvert(\vec{a}_2-\vec{a}_1)\times\vec{b}\rvert}{\lvert\vec{b}\rvert}$ | cross with common direction | parallel pairs |
| Plane: $\vec{r}\cdot\vec{n} = d$; $ax+by+cz = d$ | normal is the ID | plane equations |
| Line-plane angle: $\sin\theta$; plane-plane: $\cos\theta$ | the trig flip | angle questions |
| Point-plane: $\frac{\lvert ax_1+by_1+cz_1-d\rvert}{\sqrt{a^2+b^2+c^2}}$ | substitute and scale | distance questions |

## Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — VECTORS AND 3D

| The mistake | What is right | The trap it sets |
|---|---|---|
| Dot-zero verdict: "parallel" | Dot zero = perpendicular; cross zero = parallel | The two verdicts swapped in options |
| Ratio triples quoted as cosines | Cosines satisfy $l^2+m^2+n^2 = 1$ | Unnormalised triples labelled "cosines" |
| Line-plane angle with cosine | It is $\sin\theta = \frac{\lvert\vec{b}\cdot\vec{n}\rvert}{\lvert\vec{b}\rvert\lvert\vec{n}\rvert}$ | The flipped-trig option |
| Cross product assumed commutative | $\vec{a}\times\vec{b} = -(\vec{b}\times\vec{a})$ | Sign-flipped vector options |

| The mistake | What is right | The trap it sets |
|---|---|---|
| Section formula weights inverted | Heavier ratio letter pulls the point | Point on the wrong side of the midpoint |
| Skew SD with the parallel-line formula | Skew needs $\vec{b}_1 \times \vec{b}_2$ | Divide-by-zero signals the wrong formula |
| Distance with dropped absolute value | The modulus kills the sign | Negative "distance" options |
| Triple product sign trusted for volume | Volume takes \|STP\| | Negative-volume distractors |

## Sixty-Second Self-Test

| Statement | Verdict |
|---|---|
| Two vectors with dot product 0 are perpendicular. | True — the defining test |
| Direction ratios $(2, -4, 6)$ and $(1, -2, 3)$ give the same line direction. | True — proportional triples |
| $\left\|\vec{a} \times \vec{b}\right\|$ can exceed $\|\vec{a}\|\|\vec{b}\|$. | False — $\sin\theta \leq 1$ caps it |
| Three coplanar vectors have zero scalar triple product. | True — flat boxes have no volume |
| The shortest distance between parallel lines uses $\vec{b}_1 \times \vec{b}_2$. | False — that cross is zero; use the parallel formula |
| A line and plane angle uses $\sin\theta$ with the direction and normal. | True — the trig flip |

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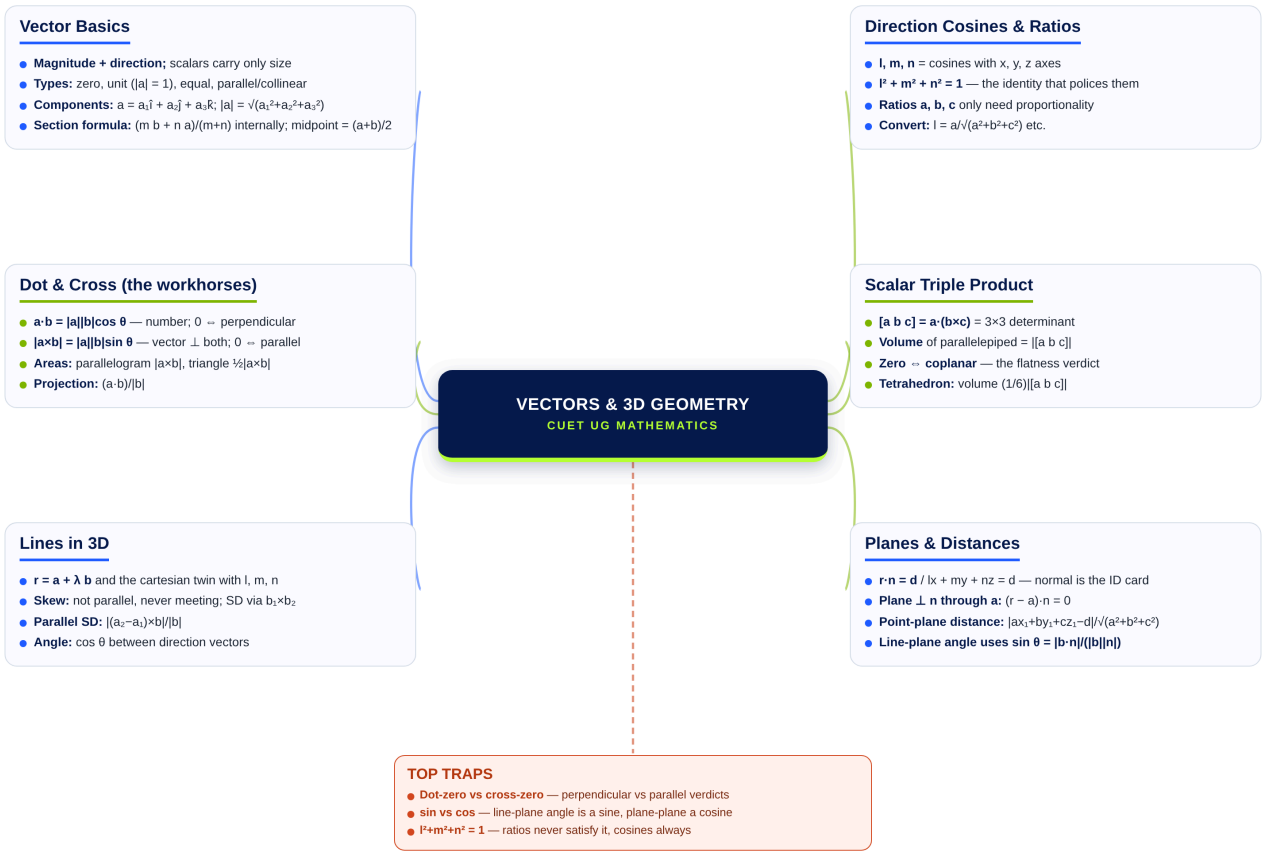



Figure 2: The chapter in one look: basics, products, lines, planes — and the traps at every junction.

## Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Name the product or formula first — then compute as little as possible.

**Q1.** The angle between $\vec{a} = \hat{\imath} + \hat{\jmath}$ and $\vec{b} = \hat{\jmath} + \hat{k}$ is:

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

**Q2.** If $\vec{a} \cdot \vec{b} = 0$ with both vectors non-zero, then $\vec{a}$ and $\vec{b}$ are:

(A) perpendicular (B) parallel
(C) equal (D) collinear

**Q3.** If $|\vec{a}| = 3$, $\left|\vec{b}\right| = 4$ and $\vec{a} \times \vec{b}$ has magnitude 12, then the angle between them is:

(A) 0 (B) $\frac{\pi}{6}$
(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

**Q4.** The area of the parallelogram with adjacent sides $\hat{\imath} + 2\hat{\jmath}$ and $2\hat{\imath} + \hat{\jmath}$ is:

**(A)** $\sqrt{14}$ **(B)** $\sqrt{10}$
**(C)** 3 **(D)** 4

**Q5.** Three vectors are coplanar if and only if their scalar triple product is:

**(A)** positive **(B)** negative
**(C)** zero **(D)** one

**Q6.** The direction cosines of the vector $2\hat{\imath} + \hat{\jmath} + 2\hat{k}$ are:

**(A)** $\frac{2}{3}, \frac{1}{3}, \frac{2}{3}$ **(B)** $2, 1, 2$
**(C)** $\frac{2}{9}, \frac{1}{9}, \frac{2}{9}$ **(D)** $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$

**Q7.** If $l, m, n$ are direction cosines of a line, then $l^2 + m^2 + n^2$ equals:

**(A)** 0 **(B)** 3
**(C)** 1 **(D)** depends on the line

**Q8.** The distance of the point $(1, 1, 1)$ from the plane $2x + y - z = 3$ is:

**(A)** 3 **(B)** $\frac{1}{\sqrt{6}}$
**(C)** $\sqrt{6}$ **(D)** 1

**Q9.** The angle between the planes $x + y + z = 1$ and $x - y = 2$ is:

**(A)** 30 degree **(B)** 90 degree
**(C)** 60 degree **(D)** 45 degree

**Q10.** The equation $\vec{r} = \vec{a} + \lambda\vec{b}$ (with $\vec{b} \neq \vec{0}$) represents:

**(A)** a plane **(B)** a point
**(C)** a sphere **(D)** a line

## Detailed Solutions

**Sol. Q1.** Magnitudes: both $\sqrt{2}$. Dot: $0 + 1 + 0 = 1$. So $\cos\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$, giving $\theta = \frac{\pi}{3}$. The 1-2-sqrt(2) triangle hides inside many dot questions.

**Answer: ( C )**

**Sol. Q2.** Dot zero with non-zero vectors means $\cos\theta = 0$, i.e. $\theta = 90°$ — perpendicularity is a dot-product verdict. Cross-zero would have said parallel.

**Answer: ( A )**

**Sol. Q3.** $\left|\vec{a}\times\vec{b}\right| = |\vec{a}|\left|\vec{b}\right|\sin\theta$, so $12 = 12\sin\theta$, $\sin\theta = 1$, $\theta = \frac{\pi}{2}$. Maximum cross magnitude happens exactly at right angles.

**Answer: ( D )**

**Sol. Q4.** Cross: $\hat{\imath}(2-0) - \hat{\jmath}(1-0) + \hat{k}(1-4) = 2\hat{\imath} - \hat{\jmath} - 3\hat{k}$, magnitude $\sqrt{4+1+9} = \sqrt{14}$. The parallelogram area is the cross magnitude itself.

**Answer: ( B )**

**Sol. Q5.** The triple product measures the box volume the three vectors span; flattening them onto one plane collapses that box, so the determinant reads exactly zero.

**Answer: ( C )**

**Sol. Q6.** Magnitude: $\sqrt{4+1+4} = 3$. Divide each ratio by 3: $\frac{2}{3}, \frac{1}{3}, \frac{2}{3}$. Option (C) divided by the sum (5-style thinking); option (B) skipped normalising entirely.

**Answer: ( A )**

**Sol. Q7.** By definition the direction cosines are the components of the unit direction vector — their squares sum to 1, always, for every line. The identity is the cosine/ratio separator.

**Answer: ( C )**

**Sol. Q8.** Substitute: $\frac{|2+1-1-3|}{\sqrt{4+1+1}} = \frac{|-1|}{\sqrt{6}} = \frac{1}{\sqrt{6}}$. Substitute fully, keep the modulus, divide by the normal's magnitude — three small steps.

**Answer: ( B )**

**Sol. Q9.** Normals $(1,1,1)$ and $(1,-1,0)$: dot $= 1-1+0 = 0$ — perpendicular normals mean the planes meet at 90°. A zero dot ended the question before any table.

**Answer: ( B )**

**Sol. Q10.** One point $\vec{a}$ plus a parameter walking along one direction $\vec{b}$ traces a straight line — the vector equation of a line. Two parameters would build a plane.

**Answer: ( D )**

## Wrong-Answer Clinic

| If you picked this | The thinking slip | The lasting fix |
|---|---|---|
| (C) in Q1 | Saw a zero component and jumped to perpendicular | Compute the dot first: 1, not 0 — then cosine |
| (B) in Q2 | Rehearsed the cross-zero verdict | Dot = alignment; cross = turning. Say both aloud |

| If you picked this | The thinking slip | The lasting fix |
|---|---|---|
| (A) in Q6 | Reported raw ratios as cosines | Divide by the magnitude; then test $l^2 + m^2 + n^2 = 1$ |
| (A) in Q8 | Dropped the modulus or mis-scaled | Full substitute, modulus, divide by $\sqrt{a^2 + b^2 + c^2}$ |
| (A) in Q9 | Used the line-line cosine on normals out of habit | Plane pairs also use cosine — but zero dot still means 90 degree |
| (B) in Q10 | One-parameter form mistaken for a plane | Count parameters: one = line, two = plane |

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